The Shape of Wisdom: How Language Models Converge on an Answer

Shailesh Rana
Independent Researcher

Abstract

When a language model answers a factual question, where does its internal state go? We asked four open models (Llama-3.1-8B, Qwen2.5-7B, Gemma-2-9B and Llama-3.2-1B) 2,256 questions whose answers recur, and recorded the state from which each answer is produced at every layer. Questions to which a model gives the same answer converge on a shared region, the answer's *place*. The answer does not become readable from the state gradually: in every model it jumps within a single layer, at a model-specific depth between 47% and 71% of the way through the network, and a step predicts the tightening of questions around their places far better than a constant per-layer contraction. Places are shared across questions: a place learned from people's citizenships reads the country the model gives for a film or a town, reading a place does not need the output directions of the answer words, and no such structure exists in an untrained network. Places are causal. After the jump, moving a wrong answer's state into the right answer's place makes the model give the right answer in 83%–97% of cases at late layers, against 2%–4% for a random move of the same length, and moves made before the jump do nothing. Finally, among wrong answers, where the state sits predicts whether the model still picks the right option in multiple choice better than the model's confidence in its reply does, in all four models.

1 Introduction

Ask a language model "What country is Aníbal Ponce a citizen of?" and, a few dozen layers later, it answers "Argentina". Between the question and the answer the model's internal state at the answering position changes at every layer. This paper asks what shape that change has. Does the state wander, drift slowly toward the answer, or arrive somewhere definite? And if it arrives somewhere, is that somewhere the same for every question with the same answer?

We find that it is. As a question passes through the network, the state moves into a region shared by all questions to which the model gives the same answer. We call this region the answer's *place*. Figure 1 shows it directly: early in Llama-3.1-8B, questions whose answers are twelve different countries form one tangled cloud; within a few layers they separate into one strand per country.

Four properties make places more than a picture. First, the answer becomes readable suddenly. In each of four models it jumps within a single layer at a model-specific depth (Section 3.1). Second, places are shared across questions: the same place is reached from different relations and wordings, and reading it does not need the output directions of the answer's own name (Section 3.2). Third, places are causal. Moving a state into a place after the jump makes the model give that place's answer, while random moves of the same length rarely do and moves before the jump are ignored (Section 3.3). Fourth, places follow the answer the model gives rather than the true answer, and this makes wrong answers legible: where a wrong answer sits predicts whether the model still recognises the right one, better than the model's confidence does (Section 3.4).

In these models and on these questions, then, the answer does not sharpen steadily layer by layer. It becomes readable at one depth, in a region shared with every other question that receives the same answer, and a confident mistake reaches the region of the answer it gives in the same way. The truth only decides which region it should have been.

2 Methods

2.1 Questions

To see whether questions with the same answer end up in the same internal state, many different questions must share each answer. We built such a set from Wikidata [36]. It contains 2,256 facts with 62 possible answers of four kinds: 24 countries, 16 cities, 12 languages[1] and 10 sports. Each answer is reached through one to three relations. A country is the answer to "which country is this person a citizen of", "which country is this town in" and "which country is this film from"; a city answers questions about birthplaces and company headquarters; a language answers questions about the language a person spoke and the original language of a film; a sport answers "which sport does this athlete play". Subjects were sampled evenly across three popularity bands (by the number of Wikipedia language editions), because models know popular facts far better than rare ones [22]; this gives a mix of facts the models know and facts they do not. Every subject has exactly one value for its relation in Wikidata, a unique English label and no namesake of the same kind.

[1] One relation–answer cell, Hindi as the language a person spoke, was dropped after a manual check found its Wikidata labels unreliable; Hindi remains as the language of films.

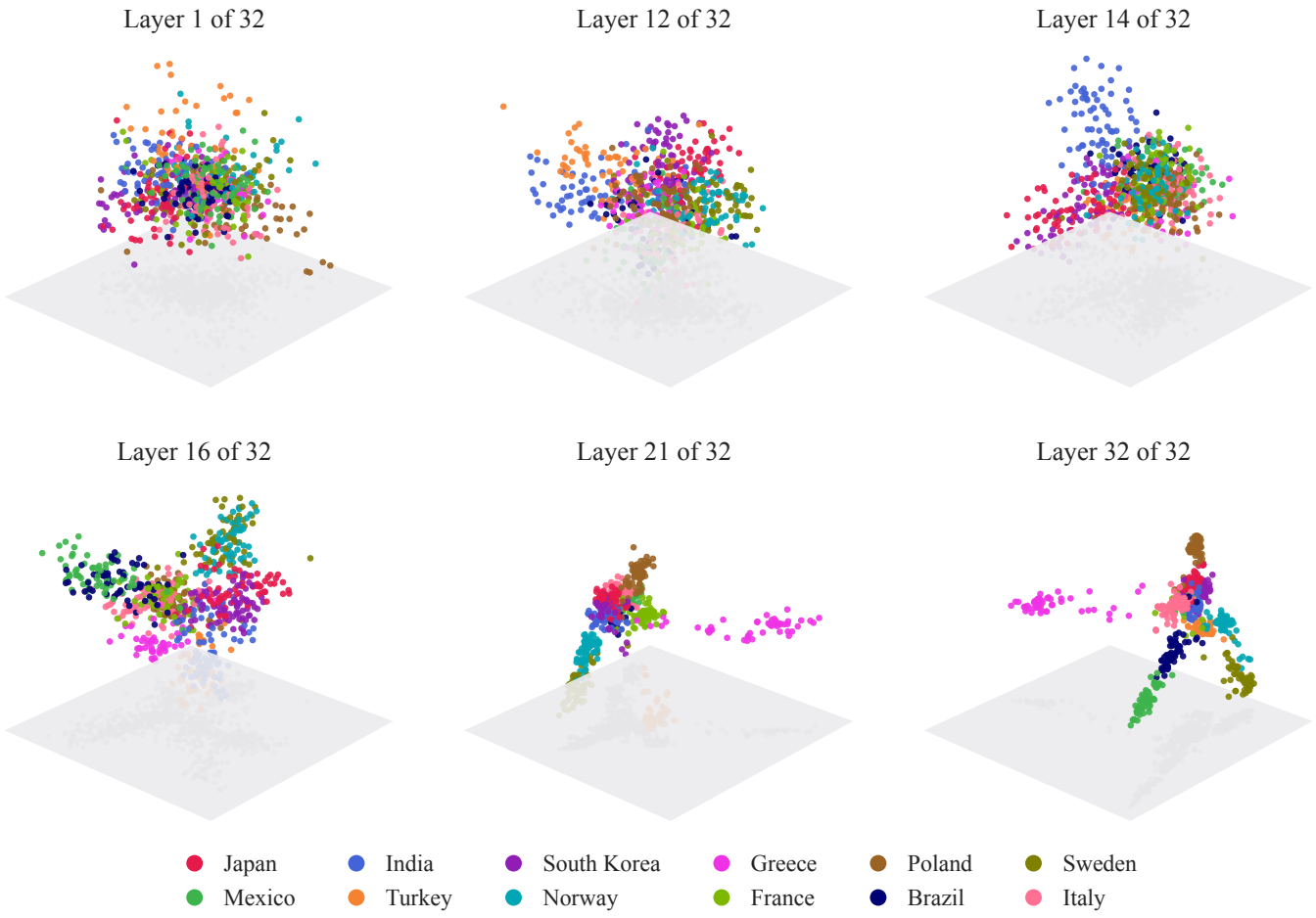


Figure 1: States of held-out questions at six layers of Llama-3.1-8B. Held-out questions about citizens, towns and films of twelve countries, as seen by Llama-3.1-8B at six layers. Each panel shows the last prompt position's state in the three directions that best separate the twelve answers among *other* (training) questions at that layer; colour is the country the model answered. Up to layer 14 the questions form one cloud. Answers become readable at layer 15; by layer 16 the questions have begun to separate, and by the last layers each country the model gives is a strand of its own. Points are scaled to a robust common range per panel; shadows are drawn on the floor.

2.2 Wordings and formats

Each fact is asked as an open question in up to three wordings: one plain template per relation ("What country is Aníbal Ponce a citizen of?"), and two rewordings in assigned styles such as a casual text message, an archivist's catalogue entry or a quiz host's question ("Hi, I'd appreciate knowing which country Aníbal Ponce is a citizen of, if possible?"). The rewordings were written by a language model that was told the subject and what to ask, but never the answer, and were then filtered automatically: each must name the subject exactly, ask a question, mention no candidate answer or its demonym, and differ from the plain wording. 4,323 rewordings passed. The plain question is also asked as four-option multiple choice in all four cyclic orders of the options, and, for a random 30% of facts, with a wrong suggestion from the user ("In which country is Ireneo Portela? I'm fairly sure it's Mexico."), which tempts the model away from an answer it knows. In total each model sees 16,294 prompts, in its own chat format.

2.3 Models and recordings

We study Llama-3.1-8B-Instruct [12], Qwen2.5-7B-Instruct [32], Gemma-2-9B-it [7], the smaller Llama-3.2-1B-Instruct, and a randomly initialised copy of Llama-3.1-8B as a control with no learned structure. The model answers greedily. At the last position of the prompt, the position from which the answer is produced, we record the residual stream at every block boundary: its value entering the first block and after each block, before the final normalisation, exactly as the model computes it in bfloat16. A free-form reply is graded only if its first line is one candidate answer (matched against hand-curated names such as "China" or "Mandarin"); replies that hedge, refuse or name two answers are set aside as unclear rather than counted as errors, and a single answer outside the candidate list counts as wrong.

2.4 Answer places

We define an answer's *place* at a given layer as the average state of all questions to which the model gave that answer. Places are defined by what the model said, not by the true

answer, so a confident mistake is grouped with the answer it names. Subjects are split in half once. Places, and every other quantity fitted to data, use only training subjects; every reported score uses only the other half. A place needs at least five prompts from at least three training subjects.

2.5 Readability and tightness

Readability asks whether the answer can be read from where the state is: for a held-out question, find the nearest place of the same answer kind and check whether it is the answer the model gave. Distances are measured after a per-layer whitening fitted on training questions (centre, keep 128 principal directions, scale each to unit variance), because a handful of very large directions in the residual stream otherwise dominate Euclidean distance. *Tightness* asks how tightly questions gather around their places, in the model's own unwhitened space: the average squared distance of held-out questions from their place, divided by the spread of the places themselves. Uncertainty comes from resampling subjects. A shuffled control repeats the readability test after permuting answers between facts.

3 Results

3.1 Onset of answer readability

Figure 2A shows how often the answer a model gives can be read from where its state is, layer by layer, for held-out questions. In every model the curve has the same shape: a low plateau, a jump, and a high plateau. The jump happens within one layer. In Llama-3.1-8B readability rises from 32% at layer 14 to 71% at layer 15; in Qwen2.5-7B from 28% to 67% between layers 19 and 20; in Gemma-2-9B from 35% to 83% between layers 25 and 26; and in Llama-3.2-1B from 35% to 74% between layers 9 and 10. Readability then keeps rising to a peak of 86%–94%. The depth of the jump differs between models, from 47% to 71% of the way through the network (Table 1), but in each it is a single event.

The early plateau is not zero. Before the jump, 20–55% of answers can be read, more than the shuffled control (6%) and more than the question's text alone (11%–18%). We think this early signal reflects surface cues in the subject's name, such as the language a name or title is in; we did not test this. Whatever its source, it does not grow toward the jump (in Qwen2.5-7B and Gemma-2-9B it fades), and the jump goes far beyond it.

Three controls rule out chance, wording and architecture as the source of the structure. Shuffling answers between facts gives 6%–8% at every layer. A classifier on the question's words alone reaches at most 18%. And an untrained Llama-3.1-8B, given the same prompts and grouped by the trained model's answers, never exceeds 8%.

Tightness. Readability depends on a whitening of the state, so a jump could in principle come from the whitening. The tightness ratio does not use one: it compares, in the model's own space, how far held-out questions sit from their place with how far places sit from each other. It falls 16- to 25-fold through the network (Figure 3). The fall is steepest at the jump, where the two layers around it account for 16%–32% of the total fall in log tightness, and continues more slowly over the next ten or so layers. We fitted two shapes to the logarithm of each model's curve, a straight line (the same contraction at every layer, the pattern found in trained classifiers [17]) and a smooth step, and compared them by how well each predicts a left-out layer. The step predicts better in every model, by a factor of 5 to 24 in error (Table 1). The step is not instantaneous: the fitted widths correspond to a rise over roughly a quarter to two fifths of the network's depth.

	Llama 8B	Qwen 7B	Gemma 9B	Llama 1B
Layers	32	28	42	16
Accuracy	63%	58%	67%	48%
Readable from (depth)	0.47	0.71	0.62	0.62
Peak readability	94%	92%	92%	86%
Error, step fit	0.025	0.017	0.051	0.034
Error, line fit	0.137	0.422	0.290	0.198
Other relation	91%	85%	90%	76%
Answer words removed	94%	90%	91%	85%

Table 1: Per-model summary. Accuracy: share of plain questions from test subjects answered correctly (unclear replies count as not correct). "Readable from": the depth at which readability first covers half the distance from chance to its peak, chosen on training subjects only. Step and line fits: mean squared error in predicting a left-out layer's log tightness. The last two rows are readability at 90% of depth.

3.2 Transfer across relations, wordings and formats

A place is reached from very different questions. Places fitted on one relation, for example people's citizenship, read the answer the model gives to the other relations with the same kind of answer, for example the country of a town or a film: at 90% of depth, 91% for Llama-3.1-8B against 94% within the same relation (Figure 2B). Places fitted on plain questions read the answers to casual or formal rewordings just as well (94%). This shows that a place does not depend on the form of the question. It does not, on its own, show that a place is more than the model's intention to say a particular word, since that intention would also be shared.

One test speaks to that. We removed from the state every direction the output layer uses to write any token of the 62 answer names, taking the final normalisation's scaling into account. Readability is unchanged (94% at 90% of depth), and removing a random subspace of the same size changes it no more. So reading a place does not need the directions that write the answer word; the place is spread over other directions too.

The multiple-choice version of each question shows where sharing stops. When the options are listed and the model must answer with a letter, we can still ask whether the multiple-choice state sits near the free-form place of the chosen option, after removing the average offset between the two formats (fit-

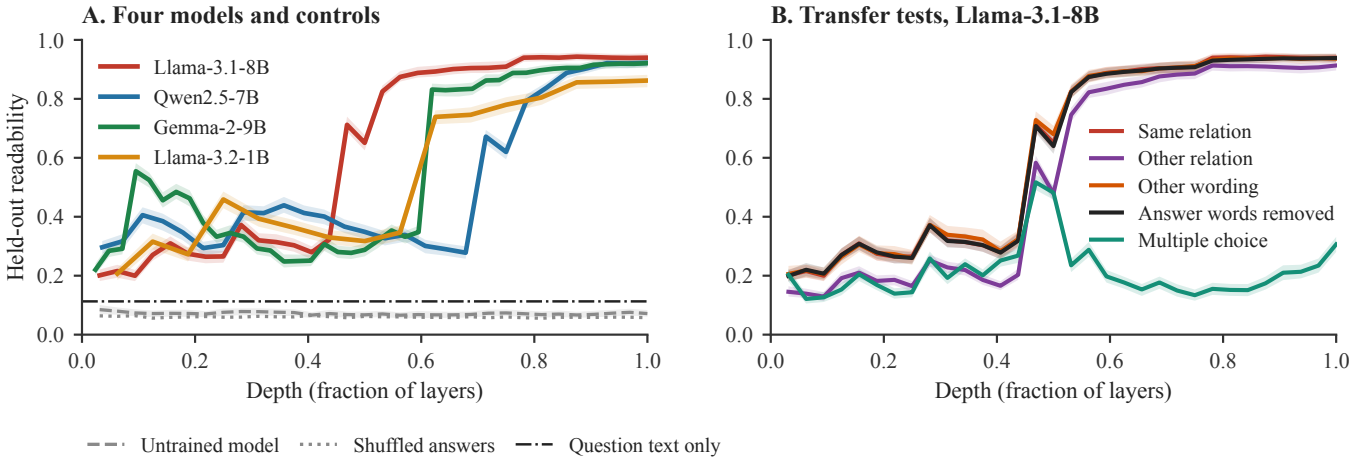


Figure 2: Held-out readability by depth. Readability is the share of held-out questions whose nearest place (fitted on training subjects, same answer kind) is the answer the model gave; bands are 95% intervals from resampling subjects. (A) All four models, with three controls: the same test on an untrained Llama-3.1-8B grouped by the trained model's answers, labels shuffled between facts, and the question's text alone. (B) Llama-3.1-8B: places fitted on one relation and tested on the others ("other relation"); fitted on plain questions and tested on rewordings ("other wording"); with the answer words' output directions removed; and tested on the states of the multiple-choice version of the same questions, after removing the average offset between the two formats.

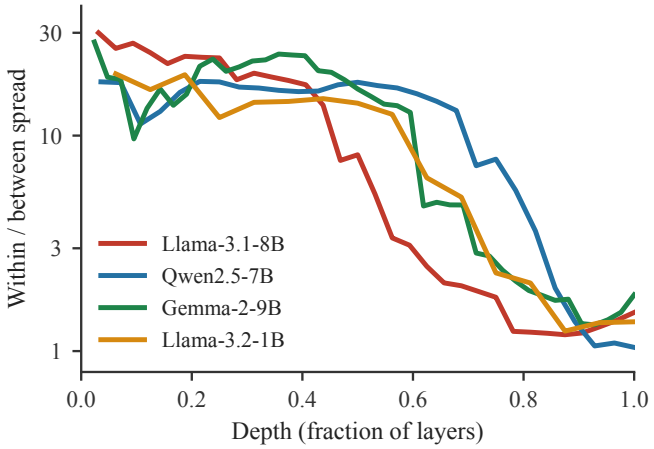


Figure 3: Tightness by depth. Average squared distance of held-out questions from their place, divided by the spread of places, in each model's own space (log scale).

ted on training subjects). At the jump it does, briefly (52% in Llama-3.1-8B, against a shuffled level near 6%), and then it fades to 21% as the state turns toward a letter. This matches the finding that models decide on the content of an option before binding it to its letter [39].

3.3 Interventions on the state

Readability shows that the answer can be read from the place; it does not show that the model uses it. To test this we moved states. For a held-out question that the model answered wrongly with an answer that has a place (121–233 questions per model), we added to its state, at one layer and at the last prompt position only, the difference between the right answer's place and the place of the answer it gave, and let the model answer again. As controls, we added a random direction of the same length, and we moved the state toward a third answer's place. Places for moving states need at least five training prompts.

The result follows the jump (Figure 4). Before a model's answers become readable, moves toward the right place do nothing: at 20% of depth they correct 0%–1% of wrong answers. After the jump they start to work, and they work more the later they are made, reaching 97% (Llama-3.1-8B), 94% (Qwen2.5-7B), 97% (Gemma-2-9B) and 83% (Llama-3.2-1B) at about 90% of depth. Random moves of the same length produce the right answer in only 2%–4% of cases. The effect is not specific to the truth: moved toward a third answer's place, the model gives that answer in 82%–95% of cases, and right answers moved toward another place are given up for it in 68%–89%. Wrong answers outside the candidate list, which have no place of their own and are moved from the average state instead, are corrected less often (58%–74% at 90% of depth, 172–264 questions per model). Within the resolution of the seven depths we tested, each model starts to follow the moves at the depth where its own answers become readable.

The moves grow with depth, from 3%–5% of the state's length at 20% of depth to 42%–49% at 90%, because places separate as depth increases. Late moves are therefore large and close to the output, and partly write the answer directly. The informative parts of the experiment are the depth at which moves start to work, the failure of random moves of the same length, and the model's readiness to follow a move to any answer.

3.4 Wrong answers

Because places are defined by what the model says, a wrong answer has a place too. At the depth where answers become readable, 45% of Llama-3.1-8B's wrong answers that name a candidate sit nearest the place of the answer it gave (58% for Qwen2.5-7B), and in Llama-3.1-8B 28% of all wrong answers sit nearest the right answer's place. This geometry carries in-

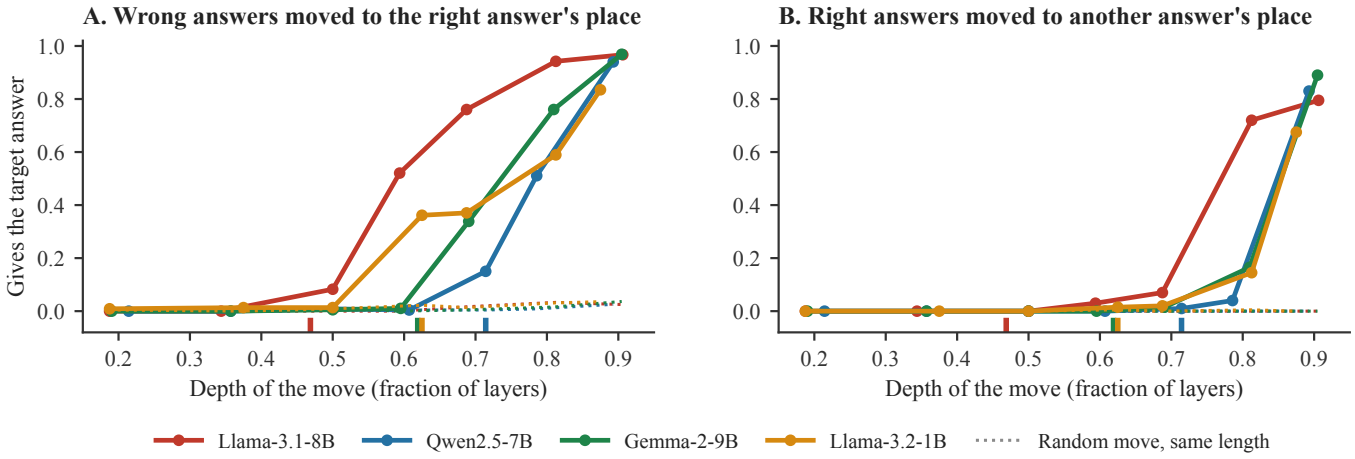


Figure 4: Effect of moving the state toward an answer's place, by the depth of the move. (A) Wrong answers moved toward the right answer's place (solid) or by a random direction of the same length (dotted). (B) Right answers moved toward a different answer's place (solid) or by a random direction of the same length (dotted). The y-axis is the share of questions for which the model then gives the answer it was moved toward. Short bars below each axis mark the depth at which each model's answers become readable. Held-out questions; places from training subjects.

formation that the reply does not. For each wrong free-form answer on a test subject, we asked whether the model still picks the right option in the multiple-choice version of the question, in at least three of its four option orders (the case for 6%–60% of wrong answers, lowest in Llama-3.2-1B). We predicted this from the model's confidence in its reply (the reply's log-probability and its length), from the geometry of the state (distance to the right place, the right place's rank among places, and the margin to the nearest place), or from both, with logistic regression and five-fold cross-validation grouped by subject. Geometry predicts recognition better than confidence in every model (Figure 5): AUC 0.75 against 0.69 for Llama-3.1-8B, and 0.73 against 0.50 for Gemma-2-9B, whose confidence carries no signal here. Adding geometry to confidence improves AUC in every model; the 95% intervals for the gain, from resampling subjects, are 0.04 to 0.13 (Llama-3.1-8B), 0.08 to 0.19 (Qwen2.5-7B), 0.15 to 0.29 (Gemma-2-9B) and 0.16 to 0.32 (Llama-3.2-1B), over 339, 463, 364 and 488 wrong answers.

The suggestion condition gives a case where we know what changed. For questions a model answers correctly when asked plainly, adding "I'm fairly sure it's ..." with a wrong candidate of the same kind sometimes changes the answer to a wrong one. Figure 6 follows the states. Where the model keeps the right answer, its state goes to the right place and stays there (99% of states nearer the right place than the suggested one at the last layer of Llama-3.1-8B). Where Llama-3.1-8B gives a wrong answer, its state first moves toward the right place at the jump, with 84% of these states nearer it than the suggested place at layer 15, and then drifts back toward the suggestion (54% at the last layer; 57 questions). The other models show the drift away from the right place late in the network, but no clear early approach.

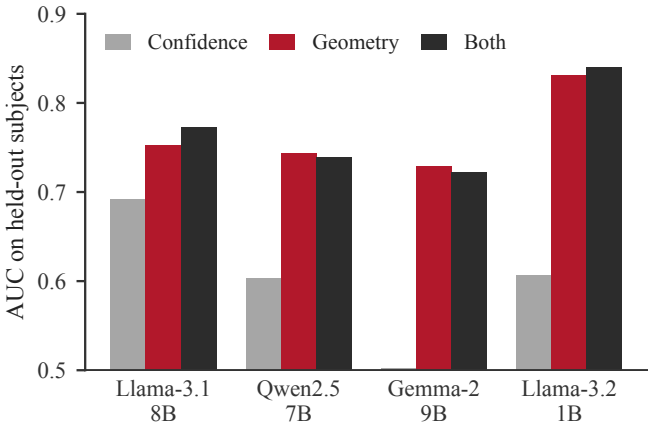


Figure 5: Predicting recognition among wrong answers. For wrong free-form answers on held-out subjects, how well the model's confidence in its reply, the geometry of its state at the onset depth, or both predict that the model still picks the right option in multiple choice (AUC, five-fold cross-validation grouped by subject; 0.5 is chance).

3.5 Given versus true answers

Two models sometimes give different answers to the same question. If places reflected the question rather than the answer each model gives, both models' states would sit near the same place. They do not: in all 12 ordered pairs of models, on questions where the two disagree, each model's state is nearer its own answer's place than the other model's answer's place in 66%–88% of cases. Consistently, places built from the model's answers read those answers far better than places built from the true answers read the truth (at 90% of depth, 94% against 75% in Llama-3.1-8B, and 90% against 65% in Qwen2.5-7B).

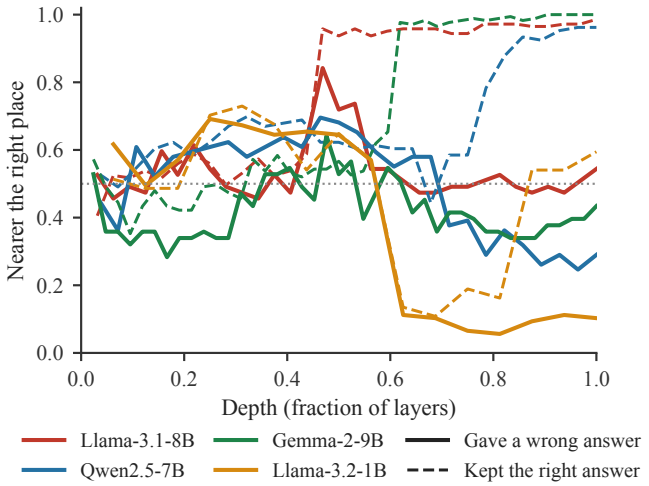


Figure 6: States under a wrong suggestion. Questions answered correctly when asked plainly, asked again with a user's wrong suggestion. Solid: the model then gave a wrong answer; dashed: it kept the right one. Y-axis: share of states nearer the right answer's place than the suggested answer's place.

4 Related work

Reading intermediate layers. Decoding intermediate states through the output layer, the logit lens [26], and its trained variant, the tuned lens [1], show predictions sharpening with depth; Future Lens [28] and Patchscopes [10] read richer content from a single hidden state. These methods read the output vocabulary. Our places are fitted from the states themselves, and reading them does not need the answer words' output directions.

How depth is used. Several studies find that layers play different roles: early layers gather and detokenise, middle layers build abstract features, and late layers refine a prediction [3, 4, 13, 20, 35]. Correct and incorrect predictions follow similar layer-wise confidence curves [19]. In factual recall, relation and then subject information move to the last position, where attention heads extract the attribute [9]; feed-forward layers act as key–value memories [8], and middle-layer feed-forward modules at the subject's last token are causally decisive [24]. Our jump in readability marks, for a whole population of questions, the depth at which each model's answer becomes available at the last position.

Shared concept spaces and representation geometry. Middle layers represent the same content in a shared space across languages and modalities [37, 41]. Concepts have structured geometry: categories form simplices [30], some features are circular [5], space and time are linearly decodable [14], counts lie on curved manifolds that attention reshapes [15], and truth has a linear direction [23]; Goodfire's Neural Geometry series [11] argues that such structure is central to understanding models. In trained classifiers, same-class representations separate from other classes by a roughly constant factor per layer [17], and language models move partway toward neural collapse (states gathering at their class means) at their top layer, further with scale [40]; the residual stream contains stable regions within which small changes leave the next-token prediction unchanged [18], and yet no two prompts map to the same state [25]. Places are consistent with both: questions can gather in the directions that carry the answer while remaining distinct in others.

What models know but do not say. Internal states often encode more than the output shows [6, 27, 33], but on knowledge benchmarks such as MMLU and ARC the gap is near zero [31], and internal states mostly reflect what the model recalls rather than what is true [2]. False few-shot demonstrations can override a correct internal answer at a critical layer [16], and users' suggestions shift answers [34]. For multiple choice, the answer letter is selected by attention in a few middle layers, and a sparse set of heads then raises its probability [38]; the winning option is decided in content space before it is bound to its letter [39]; and models carry a prior over letters [42].

Steering. Adding a direction to the residual stream changes behaviour [21, 29]. Our pushes are of this kind, but each one moves a single question's state from one measured place to another, at one layer, and is compared with random pushes of the same length.

5 Limitations

The questions are factual and short, and their answers come from a closed set of 62 countries, cities, languages and sports; places for open-ended answers, multi-step reasoning or long contexts may look different. We studied four instruction-tuned models between 1 and 9 billion parameters, one prompt format each, and greedy answers. Readability uses a whitening of the state; the tightness ratio does not, and it falls fastest at the same depth, but both measure where states sit, not the computation that moves them there. The step fit compares two simple shapes; it shows that a sudden change describes the data much better than a constant contraction, not that the change happens within a single layer. Moves at late layers partly write the answer directly, so the strongest causal evidence is the depth at which moves start to work and the failure of random moves. Wikidata answers are a reference, not ground truth; we removed one unreliable relation–answer pair after a manual check, and a small number of wrong labels surely remain. The rewordings were written by a language model and filtered automatically.

6 Conclusion

We recorded the state from which four language models answer short factual questions and followed it through every layer. In all four, the answer the model will give becomes readable from the state within a single layer, at a depth that differs between models but is fixed within each. From that

depth on, questions with the same answer gather in a shared region, reached alike from different relations and wordings and readable without the directions that write the answer's name. Moving a state into a region after that depth makes the model give the region's answer, and moving it before that depth does nothing. Regions follow the answer the model gives rather than the true one, so a wrong answer sits somewhere specific, and where it sits predicts whether the model still recognises the right answer better than its confidence does. Whether the same picture holds for open-ended answers, for reasoning over several steps and for larger models is the natural next question.

Code and data. Code, the dataset and all derived results: https://github.com/gut-puncture/The-Shape-of-Wisdom.

Acknowledgements. This work was inspired by Goodfire's Neural Geometry series [11]. The questions come from Wikidata [36]. Language models (Claude, GPT and Fable) helped with literature review, code review and drafting; all experiments, numbers and claims were checked against the recorded data.

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